

	A	B	C	D	E	F	G	H	I
1	In a sample of 200 employees, number of smoker found is 75.								
2	Construct i) 99% C.I. for population proportion of smoker.								
3		ii) 90% C.I. for population proportion.							
4		iii) Confidence interval such that population proportion of smoker certainly lie.							
5									
6	Solution:-Here, we have								
7			Sample Size (n) =	200					
8			No of smoker(x) =	75					
9			Sample proportion (p) =	0.375	=D8/D7				
10			q =	0.625	=1-D9				
11			S.E(p) =	0.03423	=SQRT(D9*D10/D7)				
12	For i) 99% Confidence interval for Population proportion								
13			Here, C.I. (1- α) =	0.99					
14			α =	0.01	=1-D13				
15			Z_{α} =	2.57583	=NORMSINV(1-D14/2)				
16			Here, Lower limit =	0.28682	=D9-D11*D15				
17			Upper Limit =	0.625	=D10-D12*D16				
18									
19	For ii) 90% Confidence interval for Population mean								
20			Here, C.I. (1- α) =	0.9					
21			α =	0.1	=1-D20				
22			Z_{α} =	1.64485	=NORMSINV(1-D21/2)				
23			Here, Lower limit =	0.31869	=D9-D11*D22				
24			Upper Limit =	0.43131	=D9+D11*D22				
25									
26	For iii) For probably lie, we should take								
27			Z_{α} =	3					
28			Here, Lower limit =	0.2723	=D9-D11*D27				
29			Upper Limit =	0.4777	=D9+D11*D27				
30									
31						Name:- Hari bashyal			